



Contents lists available at ScienceDirect

Journal of Second Language Writing

journal homepage: www.elsevier.com/locate/jslw

Book Review

Understanding development and proficiency in writing: Quantitative corpus linguistic approaches,
Philip Durrant, Mark Brenchley, Lee McCallum (Eds.). Cambridge University Press (2021). pp. x+243.
Hardback: \$110.00; Paperback: \$29.99; E-book: \$24.00

With developments in corpus linguistics and natural language processing, data-driven, quantitative research into learner writing has made much progress in describing the language use of language learners. By means of corpora, researchers are in a position to extract linguistic structures, explore the underlying principles of language systems, and have a deeper understanding of language learners' linguistic features. In this setting, Philip Durrant, Mark Brenchley, and Lee McCallum's *Understanding Development and Proficiency in Writing* (henceforth *UDPW*) reviews approximately 250 studies published over the last seventy years in L1 and L2 English writing following a quantitative corpus linguistics (QCL) approach. It provides a comprehensive overview and fully reflects the characteristics and development of the application of a QCL approach in writing.

This book contains seven chapters and can be divided into three parts. Part one consists of Chapters 1 and 2, which address the research background and the QCL approach. Part two, comprising Chapters 3–6, then constitutes the main body of the book and systematically reviews the empirical studies of learners' writing along four dimensions. Finally, part three, Chapter 7, concludes the whole book and proposes prospects for future studies. In the remainder of this book review, we first introduce each chapter and then briefly comment on several aspects.

In part one, Chapter 1 introduces the background, aim, and scope of the monograph. The authors divide the concept of writing development into two main axes of time and quality. The authors especially emphasize that they regard the improvement of language competence as effectively using various linguistic structures rather than mechanically raising the accuracy of written language. Chapter 2 presents the theoretical framework and the methods used. The authors, nevertheless, do not see the QCL approach as the only approach. They just regard it as one particular lens for certain research topics, and also point out its deficiencies and limitations.

Part two is the main body of the monograph as noted above. Durrant et al. collect and review the L1 and L2 research over the past 70 years, dividing it into four groups along the dimensions of syntax (Chapter 3), vocabulary (Chapter 4), formulaic language (Chapter 5), and cohesion (Chapter 6). Although each topic constitutes a separate chapter, they all concern the same single phenomenon, that is, the use of language in writing, as the authors mentioned in Section 1.1 (p. 6). In what follows, we summarize each chapter of this part in turn.

Chapter 3 is the lengthiest among all chapters, touching upon a large number of indicators of syntax development. The authors distinguish generic indices from granular-oriented measures as a basis for the development of syntax. The macro, generic indices have a dominant position in measuring syntactic complexity. As for the granular-oriented measures, Sections 3.3.3–3.3.7 have summarized a variety of them, which are further categorized into more elaborate subclasses. The whole chapter unfolds gradually and combines both macro- and micro-levels organically.

Chapter 4 discusses writing development from the perspective of vocabulary. Learners' vocabulary can be described either from the angle of quantity, i.e., the breadth of vocabulary knowledge, or from quality, that is, the depth of vocabulary knowledge. The authors adopt Read's (2000) framework of classification, and distinguish between three dimensions (lexical diversity, lexical sophistication, and lexical density), yet excluding the dimension of errors. Remarkably, although the existing lexical sophistication measures concern depth of vocabulary knowledge, they are still considered to be "at an extremely shallow level of depth" (p. 142). In addition, the present studies on semantic measures are done primarily through a manual analysis, whereas a deeper investigation requires more sophisticated research methods. As a result, the authors note the great prospects of developing new measures in this respect.

Chapter 5 analyzes the studies of formulaic language that is of importance in language acquisition and syntactic development. The whole chapter focuses on L2 research, yet the authors also present an L1 study by their own team. They primarily summarize the empirical studies following the Frequency-Based Approaches to Formulaic Language along two directions: one is to investigate how

¹ Tsy Yih is the transliteration of the name of the reviewer in his mother tongue, Shanghai Wu Chinese. He is also known as ZI YE in Mandarin pinyin

often a particular combination of words appears, and the other is to survey the strength of association between its components. They also review three quantitative methods in the L2 studies concerning formulaic language, where the first two methods are objective, automated procedures for identifying targeted word combinations in learners' texts or assignments, and the third one relies on researchers' subjective judgements.

Whereas Chapters 3–5 concentrate more on the relation between the indicators and language development, Chapter 6 surveys how connectedness is related to the latter on a discourse level. The authors first differentiate *coherence* and *cohesion* for the reader, with the former being a mental construct, and the latter referring to linguistic features in a text. This book pays more attention to cohesion, the research of which falls into three models – Halliday and Hasan (1976), Graesser and McNamara (2011), and the Tool for the Automatic Analysis of Cohesion (Crossley, Kyle, & McNamara, 2016), and the authors introduce each of them in turn. However, through reviewing the literature, the authors conclude that measures regarding cohesion have a long way to go.

Chapter 7 summarizes the findings and concludes with both theoretical and methodological aspects. Although there are many indicators to measure writing development, because writing is a complex process affected by various intra- and extra-linguistic factors, only a few of them are consistent across both L1 and L2 studies. For future studies, the authors offer several suggestions. The first is to keep in mind theorization. Researchers are encouraged to clearly define the indices, make transparent the links between measures and constructs, and pay attention to the potential new measures. Next, researchers should broaden attention to contextual factors, including but not limited to genres, topics, native languages, as well as extend to more participant groups. The third is to integrate with other methods, such as qualitative interviews, testing, and psycholinguistic experiments, to complement each other for the better study of writing development.

UDPW thus presents a vast collection of research on writing development that follows the QCL approach. The monograph shows systematically and comprehensively how quantitative methods are employed to study L1 and L2 acquisition of written English. There are several apparent advantages of this monograph. On the one hand, with regard to the content, we believe the authors have successfully shown the merits of the QCL approach and the reader is likely to be convinced to adopt such an approach after reading this book. Because language regularities are derived from the language use, the QCL approach enables us to find language regularities of the language used by learners from authentic language materials and provides us with a quantitative dimension of perceiving language learners' writing development, which helps researchers observe learners' language use clearly and objectively. On the other hand, regarding the reader-friendliness of the book, three points are worth noticing. First, the writing logic on the whole is clear and the content abundant. In each chapter of the main body the authors present the theoretical framework, review of studies, and methodological issues, covering both theoretical and applicative significance. Additionally, the quantitative measures of writing development are widely represented, from generic to fine-grained indices. Finally, at the end of each chapter, the authors conclude and comment from a neutral perspective. Yet, although this book has investigated previous work on writing development, we would like to supplement and delve into some points, with key attention paid to several new methods applied to this area that could possibly be addressed in a future edition.

First, the authors invariably talk about *second language* and L2 when it comes to the L2 research, without mentioning the important concept of *interlanguage* at all. As discussed many decades ago, interlanguage is a kind of natural language produced by L2 learners, and is also a language system per se (Selinker, 1972). From the very beginning of L2 development, it has the same inherent properties as other human languages, but shows independent features from the native language and the target language (Adjemian, 1976; Tarone, 1979). It manifests strong systematicity and internal consistency at any stage, has its own phonological, syntactic, and lexical rules, and is also constrained by working memory. We see the neglect of interlanguage theory in the present book as an important omission.

Second, in part one, the authors concur that the study of language development should be done from a holistic, systemic way. For example, in Chapter 2, they do emphasize that researchers should investigate the syntactic features in a syntactical way instead of splitting them into different parts, and that linguistic features do not act in isolation from each other. However, they fail to include specific holistic methods in writing development research other than Biber's (1988) multidimensional analysis (MDA). Another such approach that has already been applied to second language writing, which we would like to supplement here, is the complex network approach. By way of this method, Jiang, Yu, and Liu (2019) investigated L2 English writing from both syntactic and lexical perspectives, and find that "there is a competition between lexicon and syntax of a single language and the developments of L2 lexical and syntactic capacity are neither linear nor unidirectional" (p. 14), reflecting the prospects and values of the complex network method in investigating learners' writing development from a holistic perspective.

Finally, as Durrant et al. noted, "each act of abstraction requires a coherent theoretical framework to allow inferences back to the real world" (p. 5). Relatively speaking, there is a lack of application of a unique systematic theoretical annotated framework to learners' corpora. Yet notably, one widely adopted framework in recent L2 writing research is dependency grammar, which focuses on analyzing the relationship between parts of language that govern other parts and those that are dependent (Hudson, 2007). The reason for adopting this framework is that there are generally many syntactic mistakes in beginners' syntax that do not abide by the phrase structure rules from time to time. Hence dependency grammar, as an alternative to phrase structure grammar, proves to be a useful tool not only in natural language engineering, but also in analyzing the learners' language (Jiang & Ouyang, 2017; Kyle, 2016; Li & Yan, 2020). Large-scale annotated treebanks make extracting the extensive part-of-speech or syntactic features possible, which lets us know more about the changes of syntactic varieties of learners on both macro- and micro-levels. Although the authors mention a handful of studies under this framework in Section 3.3.4 (pp. 83–84), Section 5.3.1 (pp. 156–157) and Section 5.4.3 (p. 169), those studies that are based on learners' annotated treebanks are not given enough attention. For example, the authors fail to include the indicator of mean dependency distance (MDD), which can be employed to explore syntactic complexity from a holistic view and a cognitive perspective. Jiang and Ouyang's (2017) finding includes that MDD can be applied as a syntactic complexity indicator of English as interlanguage. Li and Yan (2020) further suggest that MDD has a universal significance in that it can measure the writing levels of

English-L2 learners with typologically different native languages. We hope that including the foregoing studies in a future edition might help make the book more informative for the reader.

In sum, the above-mentioned minor deficiencies of *UDPW* will not overshadow its merits. This book provides a holistic picture for those who are new to and interested in the field. It contains solid theoretical foundations on the one hand and abundant empirical case studies on the other, allowing the reader to see new opportunities and methodological changes in writing development research. Finally, Durrant et al. modestly acknowledge the questions in the existing QCL approach and propose prospects for future research in the book. Such a meticulous approach would no doubt lead to a productive development of this method, which can be expected to be usefully applied to the areas of writing development and language acquisition.

References

- Adjemian, C. (1976). On the nature of interlanguage systems. *Language Learning*, 26(2), 297–320.
- Biber, D. (1988). *Variation across speech and writing*. Cambridge: Cambridge University Press.
- Crossley, S. A., Kyle, K., & McNamara, D. (2016). The development and use of cohesive devices in L2 writing and their relations to judgments of essay quality. *Journal of Second Language Writing*, 32, 1–16.
- Graesser, A. C., & McNamara, D. (2011). Computational analyses of multilevel discourse comprehension. *Topics in Cognitive Science*, 3, 371–398.
- Halliday, M. A. K., & Hasan, R. (1976). *Cohesion in english*. Harlow: Longman.
- Hudson, R. (2007). *Language networks: The new word grammar*. Oxford University Press.
- Jiang, J., & Ouyang, J. (2017). Dependency distance: A new perspective on the syntactic development in second language acquisition: Comment on “Dependency distance: A new perspective on syntactic patterns in natural language” by Haitao Liu et al. *Physics of Life Reviews*, 21, 209–210.
- Jiang, J., Yu, W., & Liu, H. (2019). Does scale-free syntactic network emerge in second language learning? *Frontiers in Psychology*, 10, 925.
- Kyle, K. (2016). *Measuring syntactic development in L2 writing: Fine grained indices of syntactic complexity and usage-based indices of syntactic sophistication* (Ph.D. dissertation). Georgia State University.
- Li, W., & Yan, J. (2020). Probability distribution of dependency distance based on a treebank of Japanese EFL learners' interlanguage. *Journal of Quantitative Linguistics*, 2, 1–15.
- Read, J. (2000). *Assessing vocabulary*. Cambridge: Cambridge University Press.
- Selinker, L. (1972). Interlanguage. *International Review of Applied Linguistics in Language Teaching*, 10(3), 209–231.
- Tarone, E. (1979). Interlanguage as chameleon. *Language Learning*, 29(1), 181–191.

Xuelin Wang^{a,b}, Tsy Yih^{c,*}

^a College of Chinese Language and Culture, Jinan University, Guangzhou, China

^b Institute of Chinese Language and Culture Education, Huaqiao University, Xiamen, China

^c School of International Studies, Zhejiang University, Hangzhou, China

* Corresponding author.

E-mail addresses: wangxuelin2019@163.com (X. Wang), yezi_leafy@hotmail.com (T. Yih).